

Lab Index for Lab Sheet 1

SN	Title	Lab Date	Signature
1.	Write a program that uses two namespaces to calculate salary for "Permanent" and "Contract" employees differently, and display the result neatly formatted using stream manipulators. <i>Permanent::calculateSalary</i> $\rightarrow$ <i>basic + 20% HRA - 10% tax</i> <i>Contract::calculateSalary</i> $\rightarrow$ <i>basic + 10% allowance (no tax)</i> <i>In main(), ask the user for the employee type and basic salary, call the matching namespace's function, and print a neatly formatted salary slip using setw, setfill, fixed, and setprecision(2) so amounts line up in two decimal places.</i>	2083-03-09	
2.	Write a program that ask the user how many students are in a class, use new to dynamically allocate an integer array of that size, and read in marks. Compute and display the highest, lowest, average, and number of students who passed (≥ 40). Then ask if the teacher wants to add more students: dynamically allocate a new, larger array, copy the old data into it, delete[] the old array, and continue. Finally delete[] the last array before exiting.	2083-03-09	
3.	Write a program with overloaded area() functions (using inline for one and a default argument for another) that calculates the area of a square, rectangle, or triangle based on the number of arguments passed. <i>[inline double area(double side) $\rightarrow$ square (1 parameter)</i> <i>double area(double length, double width) $\rightarrow$ rectangle (2 parameters)</i> <i>double area(int base, double int, double factor = 0.5) $\rightarrow$ triangle (3 parameters; factor defaults to 0.5)]</i>	2083-03-09	
4.	Write a program that swaps two integers using a pointer-based function and a reference-based function, and shows why each does or doesn't change the original values in main(). <i>[swapByValue(int a, int b): ordinary value parameters</i> <i>swapByPointer(int *a, int *b): pointers are passed by value, but dereferencing them swaps the actual caller variables</i> <i>swapByReference(int &a, int &b): reference parameters]</i>	2083-03-09	
5.	Write a program that passes an array to a function to find its sum, maximum, and minimum, and another function to check whether the array is sorted.	2083-03-09	

Lab Index for Lab Sheet 2

SN	Title	Lab Date	Signature
1.	Write a program that defines a class "Student" with private data members for roll number, name, and marks obtained in 5 subjects. Use a parameterized constructor to initialize the roll number and name, and a member function to input the marks. Compute the total, percentage, and grade, then display the complete result.	2083-03-19	
2.	Write a program that defines a class "Employee" with private data members empId, name, basicSalary, hra, da, grossSalary, tax, and netSalary. Use a parameterized constructor to initialize empId, name, and basicSalary, and member functions to calculate the gross salary and net salary.	2083-03-19	
3.	Write a program that defines a class "Room" with private data members length, breadth, and height (in feet). Implement a default constructor, a parameterized constructor, and a copy constructor. Write member functions to compute the area of the floor and the volume of the room, and a function to display the room's dimensions along with the computed area and volume.	2083-03-19	
4.	Write a program that defines a class "ComplexNumber" with private data members real and imag. Implement a default constructor and a parameterized constructor. Write member functions to add and subtract two complex numbers, where one object is passed as a function argument and a new object is returned as the result. Display the complex numbers in the form a + bi.	2083-03-19	
5.	Write a program that defines a class "Counter" with a private static data member count that keeps track of the total number of objects created. Use a constructor that increments count every time an object is created, and a destructor that decrements count every time an object is destroyed. Implement a static member function to return the current value of count.	2083-03-19	